



Managing the Lock Table

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Managing the Lock Table

This topic discusses tools for viewing and managing the [lock table](#) in InterSystems products. (Also see [Monitoring Locks](#).)

1 Available Tools for Managing the Lock Table

You may find it necessary to view [locks](#) and (occasionally) remove them. InterSystems provides the following tools for this:

- The **Locks** page of the Management Portal. Here you can [view locks](#) and [remove locks](#).
- The [^LOCKTAB utility](#).
- The %SYS.LockQuery class, which lets you read lock table information.
- The SYS.Lock class, which is available in the %SYS namespace

For information on the latter two classes, see the class reference.

2 Viewing Locks in the Management Portal

You can view all of the locks currently held or requested (waiting) system-wide using the Management Portal. From the Management Portal, select **System Operation**, select **Locks**, then select **View Locks**. The **View Locks** window displays a list of locks (and lock requests) in alphabetical order by directory (**Directory**) and within each directory in collation sequence by lock name (**Reference**). Each lock is identified by its process id (**Owner**), displays the user name that the operating system gave to the process when it was created (**OS User Name**), and has a **ModeCount** (lock mode and lock increment count). You may need to use the **Refresh** icon to view the most current list of locks and lock requests. For further details on this interface see [Monitoring Locks](#).

ModeCount can indicate a held lock by a specific **Owner** process on a specific **Reference**. The following are examples of **ModeCount** values for held locks:

ModeCount	Description
Exclusive	An exclusive lock, non-escalating (LOCK +^a(1))
Shared	A shared lock, non-escalating (LOCK +^a(1)#"S")
Exclusive_e	An exclusive escalating lock (LOCK +^a(1)#"E")
Exclusive_E	An exclusive/shared escalated lock, as a result of the number of locks on various nodes of this global exceeding the lock threshold
Shared_e	A shared escalating lock (LOCK +^a(1)#"SE")
Shared_E	A shared escalated lock, as a result of the number of locks on various nodes of this global exceeding the lock threshold

ModeCount	Description
Exclusive->Delock	An exclusive lock in a delock state. The lock has been unlocked, but release of the lock is deferred until the end of the current transaction. This can be caused by either a standard unlock (LOCK -^a(1)) or a deferred unlock (LOCK -^a(1)#"D").
Exclusive,Shared	Both a shared lock and an exclusive lock (applied in any order). Can also specify escalating locks; for example, Exclusive_e,Shared_e
Exclusive/n	An incremented exclusive lock (LOCK +^a(1) issued <i>n</i> times). If the lock count is 1, no count is shown (but see below). Can also specify an incrementing shared lock; for example, Shared/2.
Exclusive/n->Delock	An incremented exclusive lock in a delock state. All of the increments of the lock have been unlocked, but release of the lock is deferred until the end of the current transaction. Within a transaction, unlocks of individual increments release those increments immediately; the lock does not go into a delock state until an unlock is issued when the lock count is 1. This ModeCount value, a incremented lock in a delock state, occurs when all prior locks are unlocked by a single operation, either by an argumentless LOCK command or a lock with no lock operation indicator (LOCK ^xyz(1)).
Exclusive/1+1e	Two exclusive locks, one non-escalating, one escalating. Increment counts are kept separately on these two types of exclusive locks. Can also specify shared locks; for example, Shared/1+1e.
Exclusive/n,Shared/m	Both a shared lock and an exclusive lock, both with integer increments.

A held lock **ModeCount** can, of course, represent any combination of shared or exclusive, escalating or non-escalating locks — with or without increments. An Exclusive lock or a Shared lock (escalating or non-escalating) can be in a Delock state.

ModeCount can indicate a process waiting for a lock, such as WaitExclusiveExact. The following are **ModeCount** values for waiting lock requests:

ModeCount	Description
WaitSharedExact	Waiting for a shared lock on exactly the same lock, either held or previously-requested: LOCK +^a(1,2)#"S" is waiting on lock ^a(1,2)
WaitExclusiveExact	Waiting for an exclusive lock on exactly the same lock, either held or previously-requested: LOCK +^a(1,2) is waiting on lock ^a(1,2)
WaitSharedParent	Waiting for a shared lock on the parent of a held or previously-requested lock: LOCK +^a(1)#"S" is waiting on lock ^a(1,2)
WaitExclusiveParent	Waiting for an exclusive lock on the parent of a held or previously-requested lock: LOCK +^a(1) is waiting on lock ^a(1,2)
WaitSharedChild	Waiting for a shared lock on the child of a held or previously-requested lock: LOCK +^a(1,2)#"S" is waiting on lock ^a(1)
WaitExclusiveChild	Waiting for an exclusive lock on the child of a held or previously-requested lock: LOCK +^a(1,2) is waiting on lock ^a(1)

ModeCount indicates the lock (or lock request) that is blocking this lock request. This is not necessarily the same as **Reference**, which specifies the currently held lock that is at the head of the lock queue on which this lock request is waiting. **Reference** does not necessarily indicate the requested lock that is immediately blocking this lock request.

ModeCount can indicate other lock status values for a specific **Owner** process on a specific **Reference**. The following are these other **ModeCount** status values:

ModeCount	Description
LockPending	An exclusive lock is pending. This status may occur while the server is in the process of granting the exclusive lock. You cannot delete a lock that is in a lock pending state.
SharePending	A shared lock is pending. This status may occur while the server is in the process of granting the shared lock. You cannot delete a lock that is in a lock pending state.
DelockPending	An unlock is pending. This status may occur while the server is in the process of unlocking a held lock. You cannot delete a lock that is in a lock pending state.
Lost	A lock was lost due to network reset.

Select **Display Owner's Routine Information** to enable the **Routine** column, which provides the name of the routine that the owner process is executing, prepended with the current line number being executed within that routine.

Select **Show SQL Options**, and then select a namespace from the **Show SQL Table Names for Namespace** list, to enable the **SQL Table Name** column. This column provides the name of the SQL table associated with each process in the selected namespace. If the process is not associated with an SQL table, this column value is empty.

The **View Locks** window cannot be used to remove locks.

3 Removing Locks in the Management Portal

Important: Rather than removing a lock, the best practice is to identify and then terminate the process that created the lock. Removing a lock can have a severe impact on the system, depending on the purpose of the lock.

To remove (delete) locks currently held on the system, go to the Management Portal, select **System Operation**, select **Locks**, then select **Manage Locks**. For the desired process (**Owner**) click either **Remove** or **Remove All Locks for Process**.

Removing a lock releases all forms of that lock: all increment levels of the lock, all exclusive, exclusive escalating, and shared versions of the lock. Removing a lock immediately causes the next lock waiting in that lock queue to be applied.

You can also remove locks using the **SYS.Lock.DeleteOneLock()** and **SYS.Lock.DeleteAllLocks()** methods.

Removing a lock requires WRITE permission. Lock removal is logged in the audit database (if enabled); it is not logged in messages.log.

4 ^LOCKTAB Utility

You can also view and delete (remove) locks using the **^LOCKTAB** utility in the %SYS namespace.

Important: Rather than removing a lock, the best practice is to identify and then terminate the process that created the lock. Removing a lock can have a severe impact on the system, depending on the purpose of the lock.

You can execute **^LOCKTAB** in either of the following forms:

- **DO ^LOCKTAB:** allows you to view and delete locks. It provides letter code commands for deleting an individual lock, deleting all locks owned by a specified process, or deleting all locks on the system.
- **DO View^LOCKTAB:** allows you to view locks. It does not provide options for deleting locks.

Note that these utility names are case-sensitive.

The following Terminal session example shows how ^LOCKTAB displays the current locks:

```
%SYS>DO ^LOCKTAB
```

```

                                Node Name: MYCOMPUTER
                                LOCK table entries at 07:22AM 01/13/2018
                                16767056 bytes usable, 16774512 bytes available.

Entry Process      X#   S# Flg   W# Item Locked
  1) 4900           1             ^["^c:\intersystems\iris\mgr\"]%SYS("CSP","Daemon")
  2) 4856           1             ^["^c:\intersystems\iris\mgr\"]ISC.LMFMON("License Monitor")
  3) 5016           1             ^["^c:\intersystems\iris\mgr\"]ISC.Monitor.System
  4) 5024           1             ^["^c:\intersystems\iris\mgr\"]TASKMGR
  5) 6796           1             ^["^c:\intersystems\iris\mgr\user\"]a(1)
  6) 6796          1e             ^["^c:\intersystems\iris\mgr\user\"]a(1,1)
  7) 6796           2             1 ^["^c:\intersystems\iris\mgr\user\"]b(1)Waiters: 3120(XC)
  8) 3120           2             ^["^c:\intersystems\iris\mgr\user\"]c(1)
  9) 2024           1             ^["^c:\intersystems\iris\mgr\user\"]d(1)

```

Command=>

In the ^LOCKTAB display, the X# column lists exclusive locks held, the S# column lists shared locks held. The X# or S# number indicates the lock increment count. An “e” suffix indicates that the lock is defined as [escalating](#). A “D” suffix indicates that the lock is in a [delock](#) state; the lock has been unlocked, but is not available to another process until the end of the current transaction. The W# column lists number of waiting lock requests. As shown in the above display, process 6796 holds an incremented shared lock ^b(1). Process 3120 has one lock request waiting this lock. The lock request is for an exclusive (X) lock on a child (C) of ^b(1).

Enter a question mark (?) at the Command=> prompt to display the help for this utility. This includes further description of how to read this display and letter code commands to delete locks (if available).

Note: You cannot delete a lock that is in a lock pending state, as indicated by the Flg column value.

Enter Q to exit the ^LOCKTAB utility.

5 See Also

- [Locking and Concurrency Control](#)
- [Details of Lock Requests and Deadlocks](#)